WATER SIM

Is it gonna flood?

Is it gonna flood?

Powered by **EO4FLOOD Datasets** from ESA and non-ESA satellites

Sentinel-1,2,3, 6, TerraSAR-X, COSMO-SkyMed, Cryosat-2, SWOT, etc.

Our Team: Multi-Disciplinary Technical & Business Expertise

Salima Bahri
PhD Solid State Nuclear Magnetic Resonance Spectroscopist | Machine Learning Engineer

Mohammed Marzouq
Spatial Computing & 3D Data Visualization Engineer | Entrepreneur

José Miguel Olvera Puentes
Climatologist, Machine Learning & Data Engineering | MSc Earth System Sciences

Amin Shakya
PhD (ongoing) in Space-borne Hydrology | MSc. Floods | Tech for Good

Rucha Nagare
UX/UI Designer | Human Centered | Problem Solver | Visual Strategist

Innovativeness & Business Viability

Housing/Real Estate
Integrate flood risk into property valuation models to address hidden risks in mortgage-backed assets.

Banking Sector
Enhance LTV and interest rate determination by improving visibility of flood risk in loan portfolios.

Insurance Companies
Improve CAT models to combat the underestimation of rare but high-impact flood events.

Governments
Optimize disaster response and infrastructure planning through improved early warning systems.

Technical Inventiveness

Advanced Machine Learning

State-of-the-art spatial-temporal modeling leveraging GCN-GRU using ESA and non-ESA satellites (Sentinel-1,2,3, 6, TerraSAR-X, COSMO-SkyMed, Cryosat-2, SWOT, etc.)

Actionable Predictions

Precise monitoring capabilities that translate into better predictions and reliable early warning signals.

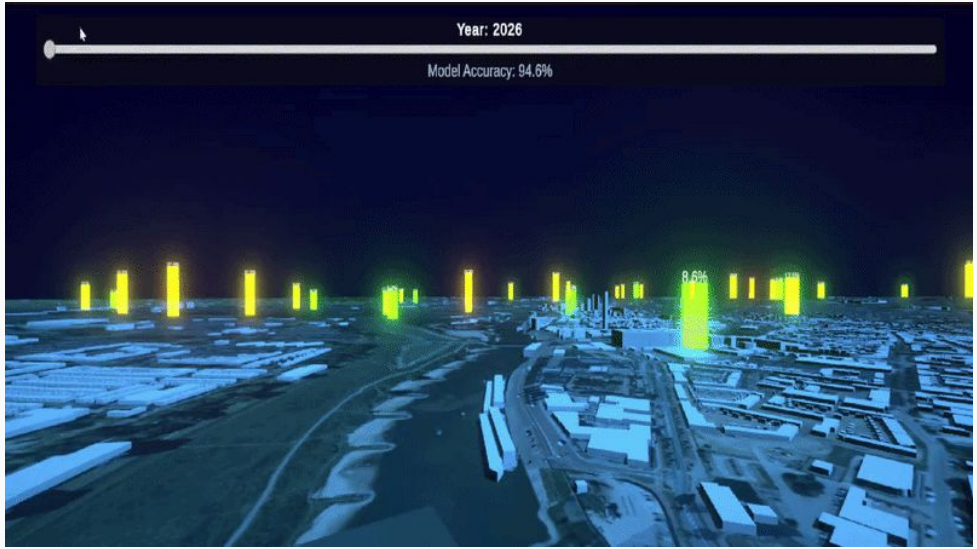
Technical Inventiveness

Spatial Computing & 3D Visualization

State-of-the-art spatial-temporal modeling leveraging GCN-GRU architectures.

Scenario Simulation

Advanced response training tools for fast and informed decision-making.



Business Model Roadmap: from high-touch B2B partnership to a scalable SaaS subscription intelligence platform.

Years 1 - 3

Product Co-Development

- Focus on ~3 key strategic clients
- Deep immersion in client needs
- Iterative product refinement

Years 4 - 6

Market Scaling

- Expansion into diverse sectors
- Standardized open subscriptions
- Proactive customer acquisition